

Energy Transformations

A player races toward the goal. Her legs power her skates to keep ahead of the defence. She shifts to the left and shoots the puck high toward the top corner.

The puck soars toward the net. Muscles in the goalie's arm contract — and she makes a fantastic glove save. The crowd roars.

Hockey players' muscle cells transform the energy stored in the food they eat into the energy needed to contract muscles for skating, shooting, and making saves. Every hockey rink is home to many energy transformations.

- Halogen lamps transform electric energy into light energy.
- Furnaces transform the energy stored in fossil fuels into thermal energy.
- Natural gas produces thermal energy to cook hot dogs and hamburgers in the concession area.

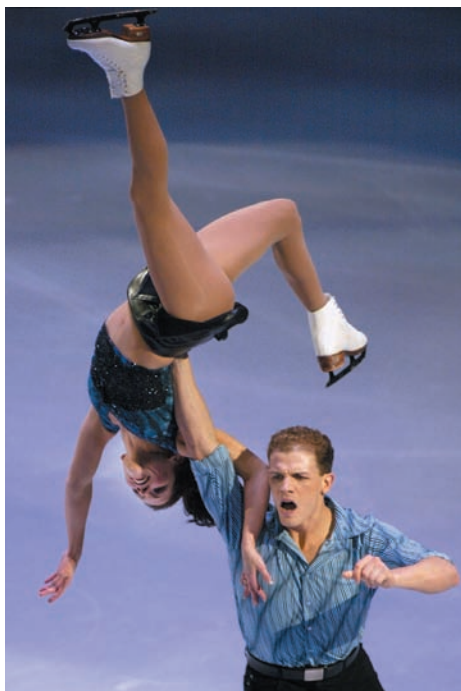
In this unit you will investigate and describe energy transformations and conservation. You will investigate machines that generate electricity from other forms of energy. You will also study the importance of balancing the amount of food energy you take in against your energy requirements throughout your life.



Energy Conversions

Getting Ready...

- List three kinds of energy used around the home.
- Much of the energy that goes into a dragster is not used to move it down the track.



As Jamie Salé and David Pelletier complete a difficult move, training and a carefully controlled diet help them transform chemical energy into motion.



A dragster converts the potential energy of fuel into the mechanical energy that drives it down the track.

Look at the photograph of Jamie Salé and David Pelletier. Stored energy from the food they have eaten allows them to complete difficult skating figures.

Fuel in the dragster works in a similar way to the food the skaters eat. That is because in energy transformations, food is a fuel. The roar of unmuffled engines (sound energy), flames exploding from exhaust pipes (sound, light, and thermal energy), and heat from engines and spinning tires (thermal energy) all point to these other energy conversions.

Consider how your life is tied to forms of energy and energy changing from one form to another.

For example, you may depend on an alarm clock changing electric energy to sound energy to wake you up in the morning. For breakfast, you probably expect electric energy to convert to thermal energy to cook your eggs and toast.

How would your life be changed if it were not possible to convert kinetic and thermal energy?

In this chapter you will take a closer look at the sources of energy you use. You will examine ways some forms of energy convert to other forms of energy. In the dragster, one kind of energy transforms to several other kinds. You will also learn how scientists account for all energy conversions.

What You Will Learn

In this chapter you will learn:

- how to identify different types of energy
- that some energy conversions absorb energy and others release it
- how energy is converted and what is lost with each energy conversion
- about the law of conservation of energy

Why It Is Important

- You need energy from food in order to fuel your body. Like your body, vehicles and machines work because they change energy from one form to another. If you understand these energy conversions, you can live economically and in good health.

Skills You Will Use

In this chapter you will:

- investigate various forms of energy conversion
- investigate temperature changes during chemical reactions
- analyze everyday technologies to identify the steps in energy conversions

5.1 Energy



Figure 5.1 Expert golfers like Tiger Woods master the way their muscles transfer energy to their club and then to the ball.

In order to get a good shot, golfers depend on energy changing from one form to another. For example, energy is converted in the golfer's muscles during the swing. When the club hits the ball, it transmits some of its energy to the ball.

Energy is the ability to do work. It can cause changes to the temperature, shape, speed, or direction of an object.

Where does your body get its energy?

- The human body converts stored (potential) energy in food into other forms of energy.
- Exercise speeds up the release of chemical energy from food stored in the muscles.

Using Energy to Do Work

You use potential and kinetic energy throughout the day.

- In the morning, you depend on potential energy when you turn on a light as you get out of bed.
- At breakfast, you count on potential energy from the foods you eat to fuel your body.

Potential energy becomes useful energy when:

- electric energy is converted to light energy to illuminate your room;
- the stored energy in food is converted to kinetic energy as you walk out the door.



Figure 5.2 These tubers have **potential energy**. Sitting at the top of a hill, they are in a position to do work.



Figure 5.3 **Kinetic energy** is energy due to motion. As they slide down the hill, tubers are doing work.



List three ways potential energy is useful to you.

5.2 Many Forms of Energy

READING check

Briefly explain how plants and animals use energy from the Sun.

We could not live without the many energy conversions that provide us with food, heat, and light. **Without energy from the Sun or solar energy, life on Earth would not exist.**

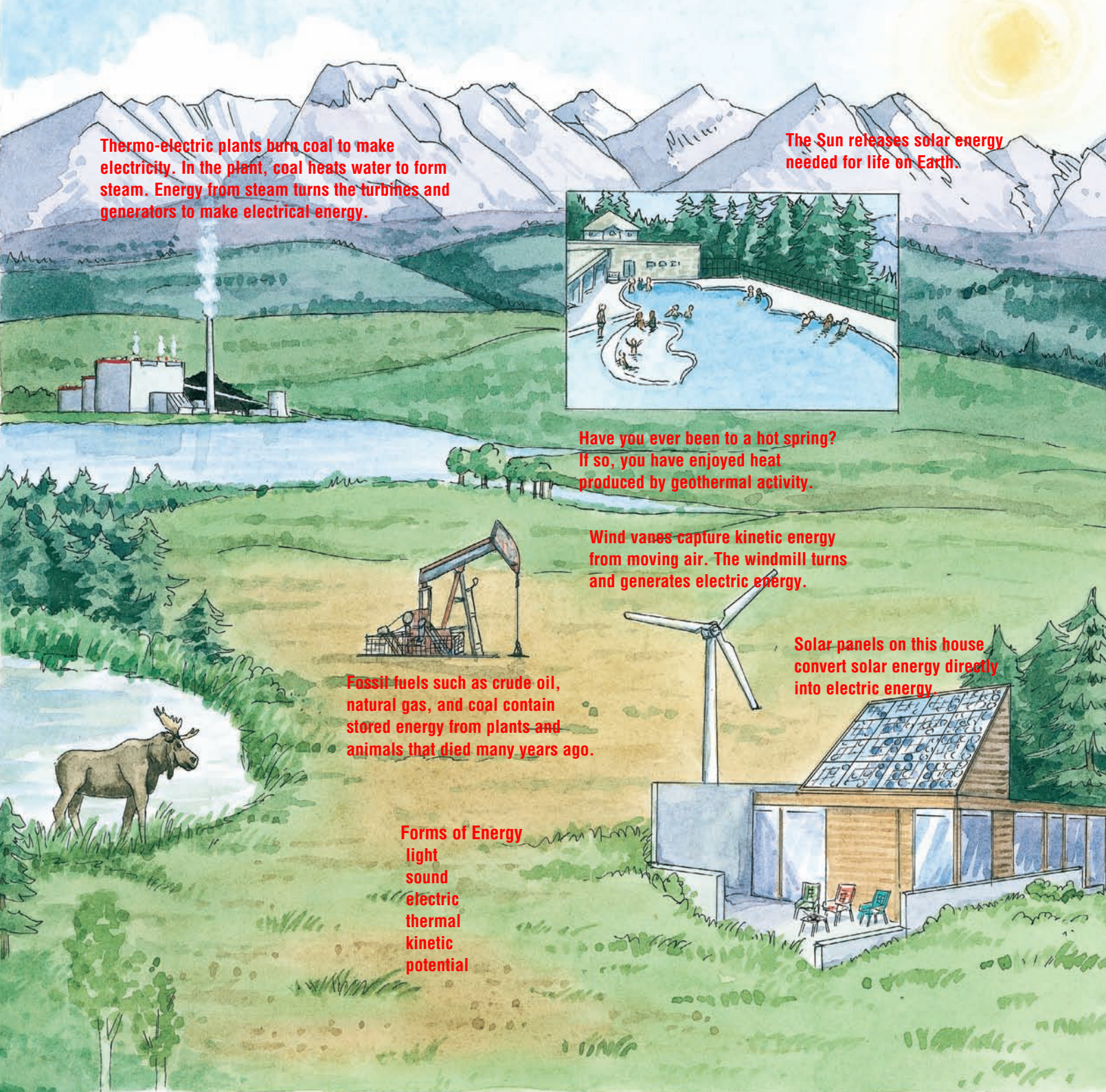
Green plants use solar energy to produce a sugar called glucose. Plants use the energy from the glucose in their own growth. Animals cannot get energy directly from the Sun. When they eat plants, they use energy from the plant to fuel their own bodies.

We use past solar energy when we burn wood. Trees use the energy produced in leaves to make new cells we call wood. A campfire produces **thermal energy**. Thermal energy provides heat. Heat is produced when the particles in substances around the flame vibrate faster and faster. Like firewood, fossil fuels such as crude oil and natural gas contain stored energy from the Sun. This energy came from plants and animals that died more than 100 million years ago.

Some homes have solar panels that convert solar energy directly to electric energy.

Other common sources of energy include **wind energy** and **geothermal energy**. Windmills capture kinetic energy from winds. As the wind vanes turn, the windmill generates electric energy.

Alberta has many sites where volcanic activity within Earth comes close to the surface. This geothermal activity is a source of thermal energy. It heats water in hot springs.



Thermo-electric plants burn coal to make electricity. In the plant, coal heats water to form steam. Energy from steam turns the turbines and generators to make electrical energy.

The Sun releases solar energy needed for life on Earth.

Have you ever been to a hot spring?
If so, you have enjoyed heat produced by geothermal activity.

Wind vanes capture kinetic energy from moving air. The windmill turns and generates electric energy.

Fossil fuels such as crude oil, natural gas, and coal contain stored energy from plants and animals that died many years ago.

Solar panels on this house convert solar energy directly into electric energy.

Forms of Energy

light
sound
electric
thermal
kinetic
potential

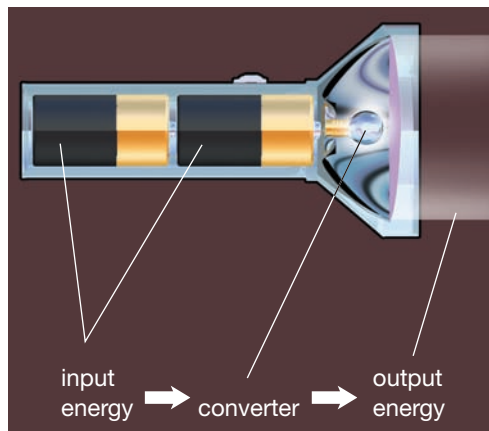
Check Your Understanding

1. List two ways the Sun's energy is converted for use.
2. Is thermal energy a form of kinetic energy? Explain.
3. What energy conversions take place from the eating of an apple to the energy used to throw a ball?

Key Terms

solar energy
thermal energy
wind energy
geothermal energy

5.3 Understanding Energy Conversion



In order to do work, energy constantly changes from one form to another. Scientists use specific terms to describe what happens during this process.

- Energy that enters the system is called **input energy**.
- The system where energy is changed from one form into another is called a **converter**.
- The energy that leaves the system is called **output energy**.

Figure 5.5 The flashlight is one example of an energy converter.

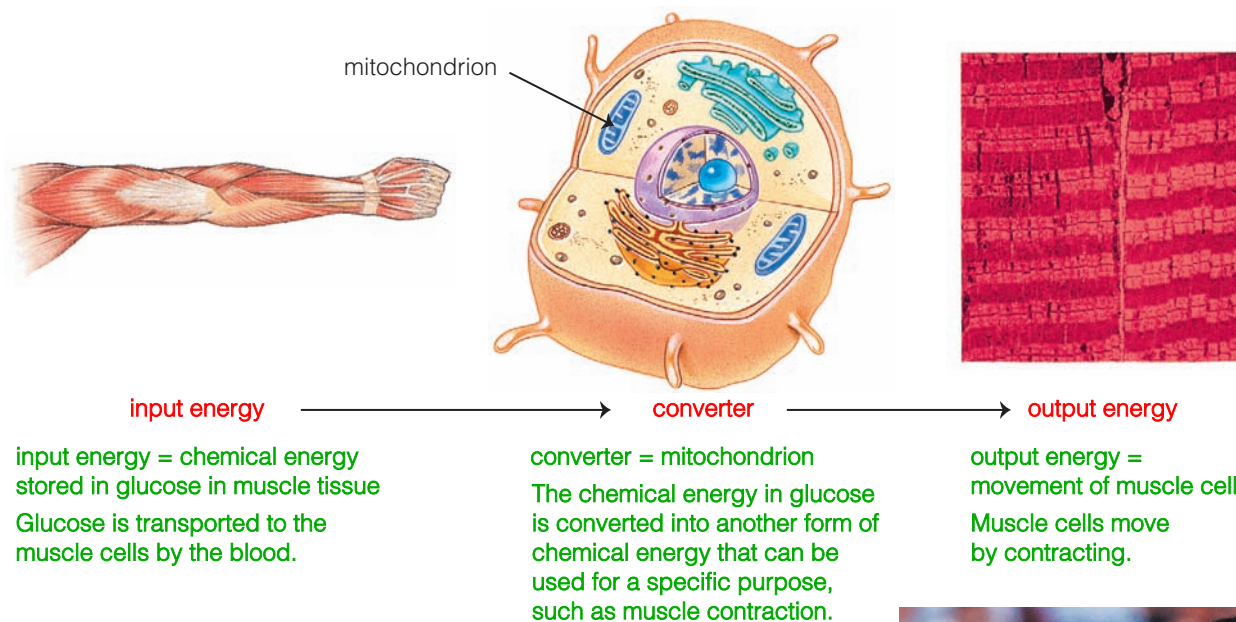


Figure 5.6 As with every energy system, the conversion of energy from one form to another is never 100 percent efficient. Some energy is released as thermal energy. You feel this when your body warms up during heavy exercise.



Chemical Conversions

The heat you feel during exercise is produced by a chemical reaction within the muscles of your body. Potential energy from food is converted to thermal energy.

Many chemical reactions produce thermal energy. In the investigation on the next page, you will learn what happens to temperature during a chemical reaction.

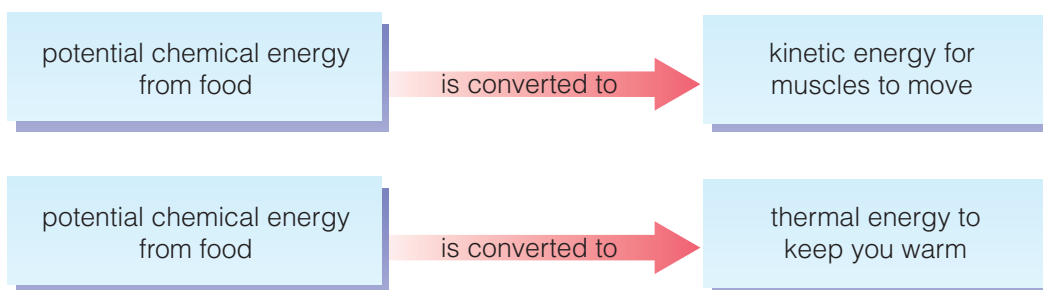


Figure 5.7 Many energy conversions occur each time you exercise. The potential energy from the food you eat converts to chemical energy as the food breaks down during digestion. This chemical energy fuels your muscles.

DidYouKnow?

About 84 percent of Alberta's electricity is produced by burning coal. Around the world, this chemical reaction has also been identified as a cause of acid deposition and climate change.



Figure 5.8 Commercial hot or cold packs such as these contain chemical potential energy. When the pack is twisted, a small container in the pack breaks and the ingredients mix. Depending on the ingredients, heat is either given off or absorbed.

INVESTIGATION 5-C

Energy Conversions Are Hard to Escape

Think About It

Is there anything you do that does not involve energy conversions? Use this activity to help you decide.

What to Do

- 1 Copy the table shown here into your notebook. Leave enough room for six empty rows.

The Converter	Input Energy	Output Energy	Is Thermal Energy Given Off?
flashlight bulb	chemical energy (battery)	light energy (bulb)	yes



B. Snowblower engine working

- 2 For each cartoon, identify the energy converter, the kind of input energy, and the kind of output energy.
- 3 Decide and record whether thermal energy is given off.



A. Person paddling



C. Drill press cutting



D. Hobby rocket taking off



E. Person trampolining



F. Person eating a sandwich

When Is Thermal Energy Wasted?

In Investigation 5–C, you learned that thermal energy is released during most activities. In many cases, this energy is waste energy. It is not used or needed for the activity. In some cases, such as during moderate physical activity, the thermal energy is not a problem. But it can become a problem on extremely hot days.

Many machines also produce excess thermal energy. The thermal energy produced by many machines often interferes with their operation. The student using the drill press must be careful to avoid overheating the drill, for example. This could result in a piece of work with burn marks.

READING check

Give two examples of waste thermal energy that might cause problems.

Key Terms

input energy
converter
output energy
chemical energy

5.4 Conservation of Energy



Figure 5.11 When we use energy to perform work such as drying our hair, do we use all of the available energy or is some wasted?

What types of energy are produced by a blow dryer?

- **Thermal energy** provides the heat to dry your hair.
- There is **kinetic energy** in the moving air.
- Some electricity is converted to **sound energy**.
- The glowing elements produce **light energy**.

You could measure the electric energy that goes into the blow dryer. If you compared it to all these forms of output energy, you would find that the two are equal.

Input energy is equal to output energy.

Scientists refer to this understanding as the **Law of Conservation of Energy**.

The Law of Conservation of Energy states that energy cannot be created or destroyed. Energy can only be changed from one form of energy to another.

READING
check ✓

Use an example to explain the Law of Conservation of Energy.

Law of Conservation of Energy

$$\boxed{\text{input energy}} = \boxed{\text{output energy}}$$

Figure 5.12 For any energy conversion, the amount of input energy is the same as the amount of output energy.

5.5 Energy Conversion Systems



Figure 5.13 The Honda Insight uses a small gasoline engine that runs at a constant speed in order to charge batteries and power electric motors that drive the car. Since the gasoline engine is small and runs at a constant speed, the car uses less fuel. As a result, it is less polluting.

Touch the hood of a car that has been for a long drive. The hood is hot. When you touch a light bulb that has been on for a while, it is hot enough to burn your fingers. Car engines and light bulbs are designed to convert energy from one form to another. As well as running a car and lighting a room, both convert some of their potential energy to other energies.

Gasoline engines produce a lot of sound and thermal energy, as well as the wanted kinetic energy. **Only about one third of the chemical potential energy of the gasoline that goes into a car engine is used to move the car.** No wonder people want cars that convert energy more efficiently.

All machines convert some energy in ways that we do not want or use. Even electric motors waste a little energy to sound and friction. Although the amount of input energy equals the amount of output energy, some of the output energy in all machines is not usable.

Law of Conservation of Energy

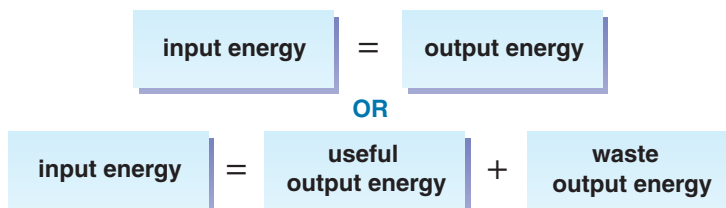


Figure 5.13 For any energy conversion, the amount of input energy is the same as the amount of output energy. Some output energy is useful. Other output energy is wasted.

DidYouKnow?

Less than one fifth of the energy that goes into a incandescent light bulb produces light!



Why is usable energy output always less than energy input?

6

Electric Energy

Getting Ready...

- What energy conversions take place inside an electric appliance?
- Explain how, with electric appliances, input energy is always greater than useful output energy.
- Why is it important to conserve electric energy?
- How can you reduce the monthly electric bill in your home?



High winds, lightning, sleet on power lines, and heavy snow are the main causes of electric power failures. In recent years, the worst instance happened when an ice storm struck Ontario, Québec, and New Brunswick. Two million people were left for days — some for weeks — without electricity.

Dairy farmers were hit hard. Without electricity, feeders and milking machines were useless. Cattle got sick and many farmers had to shoot most of their herd. Dead cows that should have been used for meat were stacked outside. With no electricity, meat packers were closed too.

Many people wondered if power would come back on before their

houses suffered major damage from frozen water pipes that burst and spilled water over floors or through ceilings.

Power outages — both the long ones and those that last only an hour — show how much we rely on electricity. Without power, can you cook your food or heat your house? How can you keep water pipes from freezing?

In this chapter you will study technologies that generate electric power. You will investigate energy conversions in electric appliances, calculate rates of energy transfer, and evaluate the efficiency of electric energy conversion systems.

at Home



What You Will Learn

In this chapter you will learn:

- about the technologies that convert other forms of energy into electric energy
- how appliances convert electric energy into other types of energy efficiently
- how to calculate the efficiency of electric devices
- how to measure electric energy and electric power
- why it is important to conserve electric energy

Why It Is Important

- Can you imagine life without light bulbs, television, cell phones, and video games? You use electric energy every day, not only for fun but for the basics like light and heat. The way we use electricity has changed the way we live. It is hard to imagine life without it.

Skills You Will Use

In this chapter you will:

- identify technologies used to generate electricity
- investigate light intensity and how solar cells produce energy
- design a model that converts the energy of flowing water into electric energy
- calculate the amount of electric power used by electric appliances
- calculate the efficiency of electric appliances

Starting Point **ACTIVITY** Where Does Electricity Come From?

Although electricity has always been around us in nature, it was only 200 years ago that people discovered how to *make* electricity. Discover for yourself how electricity is made.

What You Need

- sandpaper
- 1.5 m thin, bare copper wire
- toilet paper tube
- electrical tape
- galvanometer
- bar magnet



What to Do

1. Use sandpaper to polish 2 cm at each end of the wire.
2. Wrap 20 coils of wire around the tube. Tape the wire in place on the tube and attach an end of the wire to each terminal of the galvanometer.
3. Move the magnet into and out of the paper tube.
4. Observe the galvanometer. Record the effect of moving the magnet. 

What Did You Discover?

1. What sign is there that something in the system has changed?
2. Which way does the needle move when the North pole goes into the tube?
3. Identify the input energy, the converter, and the output energy in this system.
4. Discuss how you might increase the amount of electric energy generated in this system. Record your group's ideas. 

6.1

Generating Electric Energy

Figure 6.1 Michael Faraday, who received little formal education. While apprenticed to a bookbinder in London, he read books on scientific subjects and experimented with electricity.



In Alberta, the energy you use to light your room at night probably came from the potential energy in a lump of coal. Energy from burning coal turns a generator, which in turn produces electricity. In this way, the potential energy in coal is converted to electric energy.

In the early 1800s, Michael Faraday investigated the magnetic field around magnets. He found that pushing a magnet through a coil of wire generated an electric current. Adding more coils or using stronger magnets produced more electric energy. Using these observations, he made the first electric **generator**. A generator is a device that makes electricity.

Find Out

ACTIVITY

Generating Electric Energy

In this activity, you will build a generator.

What You Need

toilet paper tube	galvanometer
sandpaper	5 m thin bare, copper wire
electrical tape	
bar magnet	

What to Do



1. Construct the same set-up you used in the Starting Point activity on page 99.
2. Move a magnet into and out of the tube. Note and record the effect on the galvanometer.
3. Repeat Step 2, moving the magnet at different speeds.
4. Repeat Steps 2 and 3. This time, hold the magnet still and move the tube back and

forth, so it covers and then uncovers the magnet. Record your observations.

5. Wrap 20 more coils of wire around the paper roll. Secure the wire with tape. Repeat Steps 2 to 4.
6. Wrap 20 more coils of wire around the paper roll. Repeat Steps 2 to 4.

What Did You Find Out?

1. What happens to the galvanometer needle when the number of turns of wire is increased?
2. Does it make any difference whether it is the magnet or the coil of wire that moves?
3. You have found how electric energy can be produced by moving a magnet through a coil of wire. Explain how you can increase the amount of electric energy generated in a system.

SKILL CHECK

Initiating and Planning

Performing and Recording

Analyzing and Interpreting

Communication and Teamwork



Figure 6.2 Canadian inventor Thomas Willson experimented with electric generators, which are sometimes called *dynamos*. He and a blacksmith built one of Canada's earliest generators that gave Hamilton, Ontario its first electric light.

Generating Electricity

A generator is one device that produces electricity. As you discovered in the Find Out activity, the two main parts of a generator are a magnet and a coil of wire. Moving a wire coil through a magnetic field produces electricity. This works when either the magnet or the wire coil move.

Making the huge amounts of electricity we need today involves gigantic generators. In the next sections, you will look at the energy conversions used to turn these generators.

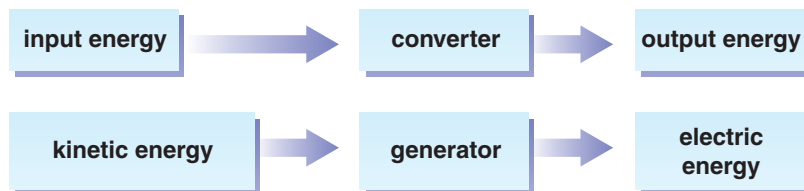


Figure 6.3 Energy conversion in a generator

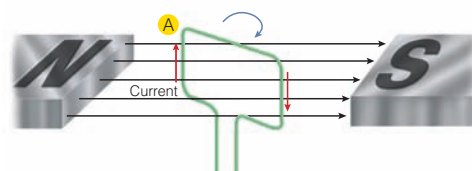


Figure 6.4 A generator consists of a magnet and a coil of wire. Spinning the coil of wire through the magnetic field is one way to generate electricity.

Try This!

Small electric motors used in model cars, trains, and airplanes can be used as generators. Connect both terminals of a small motor with a short, thin wire. Connect the wire to a galvanometer. Spin the motor. What happens to the needle of the galvanometer?

Check Your Understanding

1. Draw and label the main parts of a generator.
2. Describe how a simple generator produces electric energy.

Key Terms

generator

6.2

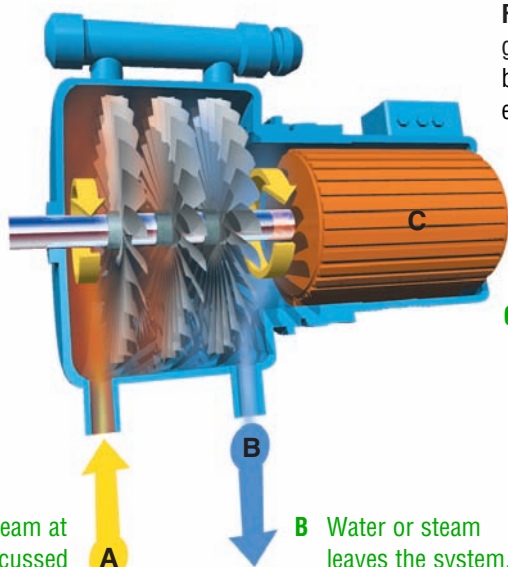
Generating and Distributing Electric Energy

READING check

List three ways electricity is generated in Canada.

Most electricity is made in a generator. Generators contain one or more turbines.

In order to make electricity, turbines need to be connected to a generator. As the blades of the turbine turn, their kinetic energy is transformed into kinetic energy in the generator. This is what produces electric energy.



A Moving water or steam at high pressure is focussed onto turbine blades and turns the machine.

B Water or steam leaves the system.

Figure 6.5 Almost all electricity is made in a **turbine**–generator combination like this. Turbines have sets of blades adjusted to the correct angle to collect the most energy from fast-moving water, steam, or wind.

C The turbine is attached to the generator. As the turbine turns, the kinetic energy of the spinning turbine is transformed into kinetic energy in the generator which, in turn, produces electric energy.



Figure 6.6A **Hydro-electric** plants convert the kinetic energy of moving water (such as water falling over a dam or waterfall) into electric energy.



Figure 6.6B **Thermo-electric** power plants convert chemical energy stored in fossil fuel. This energy is converted to thermal energy, then to kinetic energy, then to electric energy.



Figure 6.6C **Thermonuclear** plants split atoms, releasing thermal energy that is converted to kinetic energy, then to electric energy.

Coal-fired Thermo-electric Generation

Coal-fired thermo-electric plants use the fuel that is cheap and plentiful in Alberta.

In modern thermal plants, smoke from the burning coal is put through *scrubbers*. These scrubbers remove some of the harmful chemicals from the smoke before it is released to the air.

Study Figure 6.8 to see how coal-fired thermo-electric plants work. Natural gas is used in a similar way. Read Table 6.1 and consider the advantages and disadvantages of using fossil fuels like coal, petroleum, and natural gas to make electricity.

Figure 6.8 In Alberta, coal is the most common fuel used to generate electric energy.

- A** Lumps of coal are broken down to a powder. This pulverized coal is blown into the combustion chamber of the furnace. Pulverized coal burns rapidly and efficiently.
- B** In the furnace, thermal energy from burning coal converts water to steam.
- C** High-pressure steam flows through pipes and onto the fan blades of the turbine. The turbine spins with kinetic energy.
- D** The spinning turbine turns the generator. The generator converts the kinetic energy into electric energy.
- E** The steam leaves the turbine and enters a cooling chamber. Steam is condensed to water in this chamber and returned to the furnace.
- F** Electricity is distributed to customers by transmission lines.

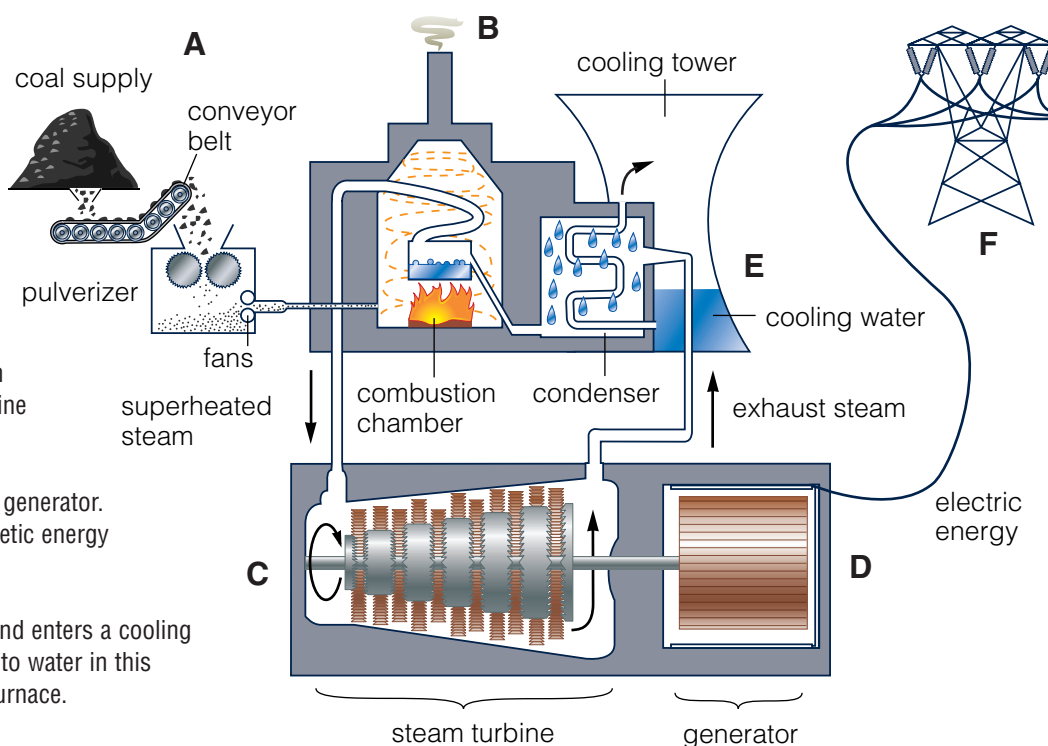


Table 6.1 Advantages and Disadvantages of Using Fossil Fuels

Advantages	Disadvantages
- Relatively inexpensive to use. - Alberta and other regions of Canada are well-supplied.	- Combustion of fossil fuels produces carbon dioxide, which can cause global climate change. - Combustion of fossil fuels may also produce sulfur dioxide, which can result in acid precipitation. - Supplies of fossil fuels are limited and cannot be renewed. - Mining and processing operations damage the environment.

Electric Energy from Water

Hydro-electric power plants produce much of the electric energy in Canada. This type of generation system uses the kinetic energy from falling water to generate electricity. Large dams are built high above the generating station.

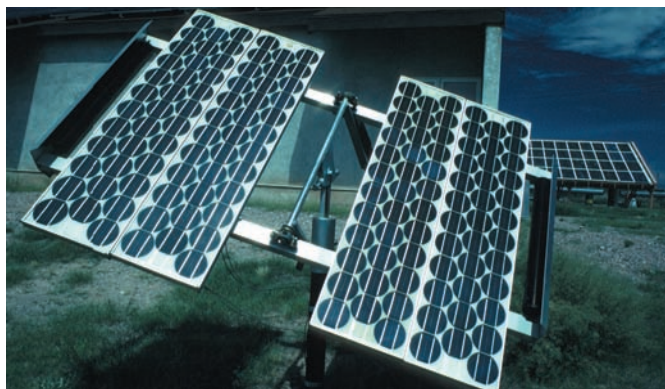
Hydro-electric power is a clean and efficient way to generate electric energy. Once the facilities are built and working, there is little environmental damage. No harmful emissions are released into the atmosphere. Generators run with little maintenance.

The main disadvantage of hydro-electric power is that the generating stations require large dams. These dams produce large lakes or reservoirs that flood land and change the ecology of an area.

READING check

List two advantages and one disadvantage to generating electric energy using the power in moving water.

There are other ways to make electricity. One of these is to have solar cells convert the Sun's energy into electric energy. This energy can be used immediately or stored in batteries. You can see solar panels powering lawn lights, calculators, electric fences, and the International Space Station.



Solar cells like these convert light energy into electric energy.



How much energy is lost as electricity moves from generating stations to your home? How is it lost?

Energy from Splitting Atoms

In a thermonuclear generating station, uranium atoms are split by **nuclear fission**. This splitting releases a tremendous amount of thermal energy. The thermal energy is used to make steam. Steam, under pressure, turns a turbine.

Study Figure 6.9 to see how thermonuclear power is created. Then, review the advantages and disadvantages of nuclear power outlined in Table 6.2.

A Uranium fuel rods are inserted into the reactor core. **Nuclear fission** begins.

B Fission reactions produce a tremendous amount of thermal energy. This energy is carried away from the reactor by a coolant.

C Coolant is taken to a tank where the thermal energy converts water into steam.

D High-pressure steam flows through pipes and is directed through the blades of a turbine. The blades move and make the shaft spin.

E The spinning turbine turns the generator. The generator converts this mechanical energy into electric energy.

F Transmission lines take electric energy to customers.

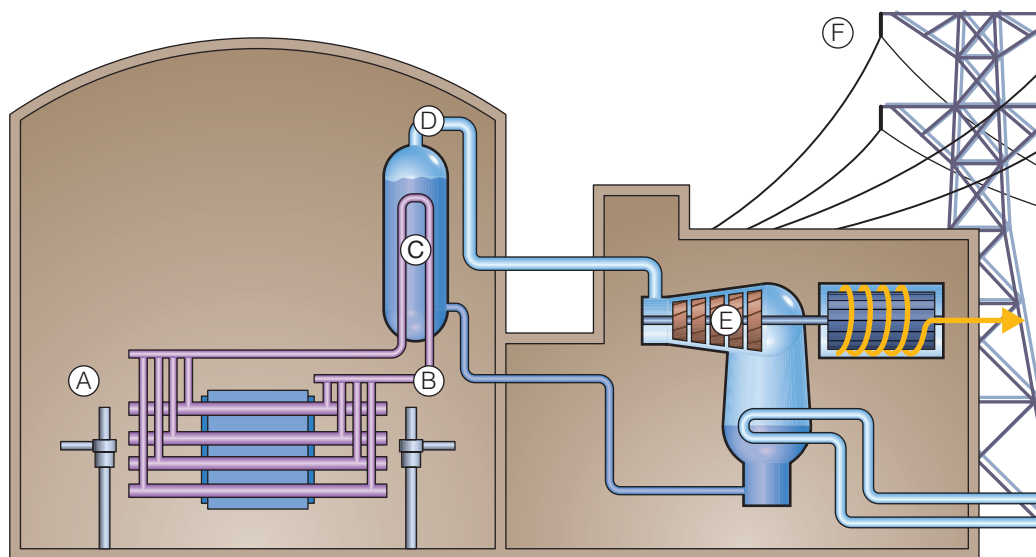


Figure 6.9 This diagram shows how a CANDU nuclear reactor produces electricity from nuclear energy. CANDU reactors were designed and built in Canada. They are used in Canada, Argentina, India, South Korea, Pakistan, China, and Romania.

Table 6.2 Advantages and Disadvantages of Using Nuclear Energy

Advantages	Disadvantages
- Reactors do not release harmful gases, such as carbon and sulfur dioxides. - A small quantity of nuclear fuel generates an enormous amount of electricity.	- Nuclear waste is dangerous to life for centuries. - Storing waste safely is expensive and dangerous. - Accidents can release radioactive materials. This material can kill plants and animals. It can also cause illnesses that show up years later.

Distributing Electricity

Electric generating plants are usually located great distances from the communities they serve. Electricity must often be transmitted hundreds of kilometres along power lines. As electricity passes through the lines, about 10 per cent is converted to thermal energy and wasted.

Thermonuclear plants, and thermo-electric plants that burn coal, are often located close to cities. This lessens power loss, but creates environmental concerns.

In Canada, long transmission lines cause other problems. Heavy storms, such as those mentioned in the opening pages of the chapter, knock down power lines.

6.3 Electric Energy and Power



Figure 6.10 Electric appliances convert electric energy into other forms of energy. What is the useful energy produced by each of these items?

DidYouKnow?

When James Watt developed steam engines, he compared the energy produced by this new technology with something everyone knew in those days. He used the term “horsepower.” One horsepower was described as the work done by one horse in one minute.

One horsepower is equal to 746 W.

Energy is the ability to do work. Work is defined as force (F) multiplied by distance (d) over which the force is applied. (Work = F x d).

Force is measured in **newtons**. When a **newton** of force is applied for one metre of distance (1 N•m), one joule of work is done. Note that the joule is the unit used in measuring work.

Electric energy is able to perform a range of work — from turning the beaters of a mixer, to spinning the blade in a table saw, to moving a freight train.

Power is the rate at which energy is transferred. Power tells how fast energy is used or produced. If a machine uses one **watt** of power, it uses one joule of energy in one second.

Because work and energy are equivalent, power is also the rate at which work is done. Work done quickly means higher power. Work done slowly means lower power.

$$1 \text{ watt} = 1 \text{ joule per second}$$

or

$$1W = \frac{1J}{s}$$



What is power?

For example, an electric hair dryer that uses 1500 joules of electric energy every second would be rated at 1500 watts. A light bulb that uses 60 joules of electric energy every second would be rated at 60 watts. The Math Connects that follow will show you how to calculate electric power.

Math CONNECT

Using a Formula to Calculate Power

A generator produces 2 J of electric energy every second. How many watts of power does it produce? Use the equation

$$\text{power} = \frac{\text{energy}}{\text{time}}$$

Steps in Solving It

1. Write down what you know.

$$\text{power} = ?$$

$$\text{energy} = 2 \text{ J}$$

$$\text{time} = 1 \text{ s}$$

$$\text{power} = \frac{\text{energy}}{\text{time}}$$

2. *What do you want to find out?*

How much power is produced.

3. *Use the formula.*

$$\text{power} = \frac{\text{energy}}{\text{time}} = \frac{2 \text{ J}}{1 \text{ s}}$$
$$= 2 \text{ J/s}$$

4. *Express the power in watts.*

The generator produces two watts of power.

Solve These

Use Steps 1 to 4 to calculate power for each of these problems.

1. An aquarium light bulb uses 7 J of electric energy in one second. How many watts of power does this light bulb consume in 3 s?
2. Speakers on a computer use 40 J of electric energy in one second. How many watts of power do the speakers consume in 60 s?

What Do You Buy from the “Power” Company?

Although electric utility companies are often called power companies, they really provide electric energy. When you pay an electricity bill, you pay for electric energy.

The consumption of electric energy is measured in watts. Usually, the watts are multiplied by the time it takes to use them. For example, if an appliance uses one watt of power every hour, it uses one **watt hour (W•h)** of energy.

$$\text{energy} = \text{power} \times \text{time}$$

$$1 \text{ watt hour (W•h)} = 1 \text{ watt} \times 1 \text{ hour}$$

READING
check ✓

What is one watt hour of energy?



Math CONNECT

Watt Hours

How much electric energy does a 1 W light bulb use if it is left on for two hours?

Steps in Solving It

1. *Write down what you know.*

$$\begin{aligned}\text{energy} &= ? \\ \text{power} &= 1 \text{ W} \\ \text{time} &= 2 \text{ h} \\ \text{energy} &= \text{power} \times \text{time}\end{aligned}$$

2. *What do you want to find out?*

How much energy is used.

3. *Use the formula.*

$$\begin{aligned}\text{energy} &= \text{power} \times \text{time} \\ &= 1 \text{ W} \times 2 \text{ h} \\ &= 2 \text{ W•h}\end{aligned}$$

4. *Express the energy in watt hours*

The light bulb uses two watt hours of energy.

Solve These

Use Steps 1 to 4 to calculate the energy used in each of these problems.

1. How much electric energy does a 100 W light bulb use if it is left on for one hour?
2. How much electric energy does a 240 W fan use if it is left on for two hours?

Kilowatt Hours

The watt hour is a small unit of energy. Electric utility companies use a unit of energy 1000 times larger. They bill customers for use of kilowatt hours ($\text{kW}\cdot\text{h}$). One **kilowatt hour ($\text{kW}\cdot\text{h}$)** is equal to 1000 watt hours.



Figure 6.11 Electric meters like this measure electric energy in kilowatt hours ($\text{kW}\cdot\text{h}$)



Calculating Kilowatt Hours

A kettle is rated at 1000 watts. This means that it uses 1000 watts for each hour it is used. During one week, it is used for two hours. How much energy does it use in kilowatt hours?

Steps in Solving It

1. Write down what you know.

energy = ?

power = 1000 W

time = 2 h

energy = power $\times$ time

2. What do you want to find out?

How much energy is used in $\text{kW}\cdot\text{h}$.

3. Use the formula.

energy = power $\times$ time

= $1000 \text{ W} \times 2 \text{ h}$

= $2000 \text{ W}\cdot\text{h}$

4. Divide by 1000 to change $\text{W}\cdot\text{h}$ to $\text{kW}\cdot\text{h}$

$$\frac{2000 \text{ watt hours}}{1000} = 2 \text{ kW}\cdot\text{h}$$

5. Summarize your results.

The kettle uses $2 \text{ kW}\cdot\text{h}$ of energy.

Solve These

Use Steps 1 to 5 to calculate the number of kilowatt hours used by each of the following appliances.

1. How much electric energy does a 1000 W table saw use when run for 3 hours?
2. How much electric energy does a 255 W television use if it is on for 4 hours?

Key Terms

newton

power

watt

watt hour

$\text{W}\cdot\text{h}$

kilowatt hour

$\text{kW}\cdot\text{h}$

Check Your Understanding

1. What units are used to measure the power of electric appliances?
2. How much energy, in kilowatt hours, is used by a 1000 watt hair dryer when it runs for one hour?
3. Why do you think electric power companies measure electric energy in kilowatt hours instead of watt hours?

6.4 Efficiency and Saving Energy

Incandescent light bulbs are designed to convert electric energy into light energy. Incandescent means “glowing with heat.” These bulbs produce light when electricity flowing through the filament makes the thin wire white hot. Some of the electric energy is converted to light. A lot of it is converted into other forms of energy.

Complete the Find Out activity to investigate the forms of energy produced by two kinds of light bulbs.

READING check

What is the major form of waste energy in an incandescent light bulb?

Find Out **ACTIVITY**

Heat Produced by Two Kinds of Light Bulbs

In this activity, you will investigate thermal energy produced by two kinds of light bulbs.

What You Need

graph paper
ruler
2 retort stands and clamps
2 thermometers
100 watt incandescent light bulb
30 watt compact fluorescent light bulb
2 wired light sockets

What to Do



1. Develop a chart to record your observations.
2. Put one light bulb in each light socket and attach them to the retort stand.
3. Clamp one thermometer so that its coloured bulb is 1 cm from the top of the incandescent light bulb. Do the same with the second thermometer and the compact fluorescent bulb.



SKILL CHECK

Initiating and Planning

☀ Performing and Recording

☀ Analyzing and Interpreting

Communication and Teamwork

A 25 watt compact fluorescent bulb produces the same amount of light as a 100 watt incandescent bulb. The compact fluorescent bulb provides the same amount of light, but uses 75 percent less energy and lasts 10 times longer than an incandescent bulb.

4. Record the temperature shown on each thermometer.
5. Turn on the lights. Record the temperatures every minute for five minutes.
6. Observe the lights from two metres away. Do they give off about the same light intensity?
7. Graph your results. Put time, in minutes, on the horizontal axis (x-axis). Put temperature on the vertical axis (y-axis).

Efficiency

Efficiency is a measure of how completely input energy is converted to accomplish the desired output energy. Incandescent bulbs and fluorescent bulbs produce about the same amount of light. Incandescent bulbs produce more heat. In this case, thermal energy is not the task. Therefore, incandescent bulbs are less efficient than fluorescent bulbs.

Efficiency can be expressed as a percentage. For example, if a motor is 80 percent efficient, 80 percent of the input energy is converted to do the intended job. Twenty percent of the input energy is wasted.

You can determine the efficiency of an electric device by dividing useful energy output by the total energy input, and then multiplying by 100. Efficiency is always expressed as a percent.

READING check

What does the term “efficiency” mean?

$$\text{Percent efficiency of an electric device} = \frac{\text{useful energy output}}{\text{total electric energy input}} \times 100$$

Math CONNECT

Calculating Efficiency

An electric kettle uses input energy of 240 000 joules to heat water. 196 000 J is actually used to heat the water. Calculate the efficiency of the kettle.

Steps in Solving It

1. Write down what you know.

useful energy output = 196 000 J

total electric energy input = 240 000 J

efficiency = ?

Percent efficiency of an electric device =

$$\frac{\text{useful energy output}}{\text{total electric energy input}} \times 100$$

2. What do you want to find out?

The efficiency of the kettle.

3. Use the formula.

Percent efficiency of an electric device =

$$\frac{\text{useful energy output}}{\text{total electric energy input}} \times 100$$

$$= \frac{196\,000\text{ J}}{240\,000\text{ J}} \times 100 = 81.6\%$$

4. Summarize your results.

The kettle is 81.6 percent efficient.

Solve These

1. An incandescent light bulb uses 30 000 J of electric energy. It emits only 900 J of useful energy. What is the efficiency of the light bulb?
2. An electric motor has 100 J of electric energy supplied to it every second. It produces 70 J of kinetic energy every second when it is providing power to a water pump. What is the efficiency of the motor?



Why is it important to conserve electric energy?

It Pays to Be Efficient

Saving energy offers many benefits. If you use less electricity, you will save money on your monthly utility bill.

Using less electricity also benefits the environment. In Alberta, most electricity is generated using fossil fuels; most of that comes from coal-burning power plants. As coal burns, it produces carbon dioxide, sulfur dioxide, and nitrous oxide gases. These waste gases are released into the atmosphere. Carbon dioxide has been linked to global climate change. Sulfur dioxide and nitrous oxides are the primary causes of acid deposition. Using less electricity means that fewer harmful emissions are released into the environment.

DidYouKnow?

It takes approximately 325 kg of coal to generate enough electric energy to keep a 100 watt light bulb on 24 hours a day for 365 days.



Figure 6.12 Turning appliances off when they are not being used saves electric energy — and the environment.

Key Terms

efficiency

Check Your Understanding

1. Why does an incandescent light bulb get so much hotter than a fluorescent light bulb?
2. State three ways to conserve electric energy at home.

Energy for Life

Getting Ready...

- Why are plants considered the basis for life?
- Do some people need more energy than others to lead a healthy life?
- How do energy requirements change throughout a person's life?



Agriculture is a major part of the Canadian economy. Fresh fruit and vegetables are important sources of energy.

Life on Earth could not exist without plants. The ability of plants to make oxygen made it possible for oxygen-breathing animals to develop on this planet. Plants convert carbon dioxide in the atmosphere into oxygen.

As well as this, they capture energy from the Sun and use it to produce food. This means that animals that eat roots, stems, leaves, and the meat of other animals all depend on plants every day.

In this chapter you will examine how plants capture energy from the Sun and convert it into potential energy. You will also see how animals use this food to obtain energy for themselves. You will examine carbohydrates, fats, and proteins, and see how each is used as a source of energy. You will also analyze how human energy needs change from infancy to old age.



Hydroponics is growing plants without soil. Plants in constant contact with water grow well under artificial light.

What You Will Learn

In this chapter you will learn:

- how plants capture energy from the Sun in the food they produce
- how the food we eat provides energy
- how our bodies use energy stored within three nutrient groups
- how the amount of energy humans need changes according to their stage of growth and activity level

Why It Is Important

- Our bodies are unable to convert solar energy to the energy we need for life. The energy we use can be traced back to the energy trapped by plants in glucose. Our energy level and our health depend on the food we consume, and good health all depends on plants. By the end of the chapter you may find yourself thanking the plants you meet.

Skills You Will Use

In this chapter you will:

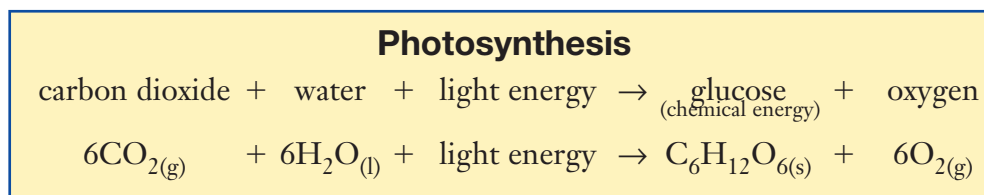
- trace the pathway of energy captured from the Sun
- evaluate the nutrient value of food choices
- investigate some foods and their energy content

7.1 Nature's Energy Conversions



Figure 7.1 Alberta's wheat fields are a major source of natural energy. Chlorophyll in the wheat plants captures the Sun's energy and converts it into chemical energy. This is stored in glucose, which is a form of carbohydrate. Scientists have copied this process in solar cells (on right), which convert the Sun's energy into electric energy.

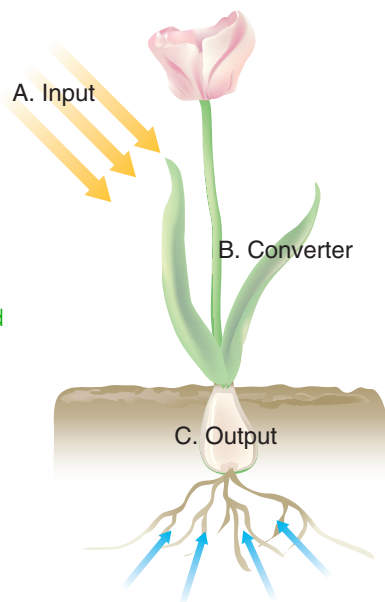
The cells of green plants have special parts that capture solar energy and convert it to energy which plants can use. These cell parts are called **chloroplasts**. Chloroplasts contain **chlorophyll**, which is a green-coloured chemical that assists in converting carbon dioxide and water into glucose and oxygen. This process is referred to as **photosynthesis**. The chemical reaction equation for photosynthesis is shown below.



A. The ingredients for photosynthesis are water from the soil, energy, energy from the sun, and carbon dioxide from the air.

B. Chlorophyll in chloroplasts traps solar energy needed to combine water and carbon dioxide into glucose and oxygen.

C. Glucose is a form of carbohydrate. It is stored in plants. Oxygen is released into the air.



Glucose, a sugar, is a simple carbohydrate that is rich in chemical energy which plants and animals can use. Plants manufacture and store this chemical potential energy in their stems, leaves, and roots. Our bodies use carbohydrates as a source of energy.

Figure 7.2 A plant captures solar energy to manufacture carbohydrates.

Cellular Respiration

Cellular respiration is a process used by all plant and animal cells to obtain energy from food. All cells use energy from cellular respiration to maintain life processes.

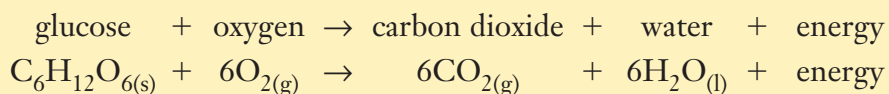
The chemical reaction equation for cellular respiration is shown below. Study the equation, then compare it to the equation for photosynthesis. What do you notice?

Glucose is the main source of energy for all cells. Cells also get energy from proteins, fats, and other food molecules.

Figure 7.3 Our bodies cannot capture energy directly from the Sun. Energy is transferred to our bodies when we eat green plants or when we eat the meat of animals that have eaten green plants.



Cellular Respiration



Cellular respiration takes place in cells in specialized organelles called the mitochondria. **Mitochondria are the “power plants” of the cell.** In order to provide fuel to this power plant, humans must consume chemical potential energy from plants and animals.

Two Energy Conversions

It takes two energy conversions for Sun power to get to you so you can do the activities you enjoy.

1. The equation below shows the energy conversion in plants.

input energy	→	converter	→	output energy
solar energy	→	chloroplasts in plant leaf	→	chemical potential energy (glucose, a form of carbohydrate)

2. Animals convert plant energy using the energy conversion that follows.

input energy	→	converter	→	output energy
chemical potential energy (glucose)	→	(mitochondria in cells)	→	muscular activity in muscle cells (kinetic energy)

Energy Conversion: Plants to Animals

During all energy conversions, a certain amount of energy becomes waste energy. During walking, for example, the body does not convert 100 percent of the chemical energy in glucose to kinetic energy. Some energy is lost. The most common type of waste energy is thermal energy lost to the surrounding environment.

Your body produces thermal energy to keep your body warm and releases the excess as waste energy. For example, have you noticed that you get warmer (that is, release more thermal energy) when you increase your physical activity? Similarly, the temperature of a cool room increases when more and more people enter. The source of heat is the thermal energy being released by the people's bodies.

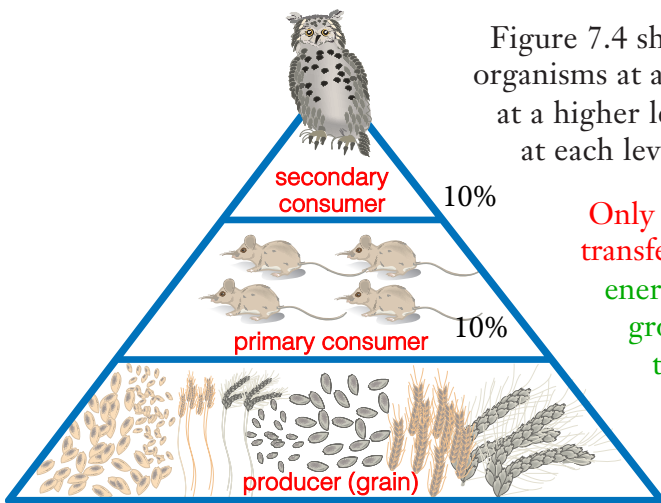


Figure 7.4 shows an **ecological pyramid**. Note that it takes many organisms at a lower level to support a small number of organisms at a higher level. This is because of the amount of waste energy at each level.

Only about 10 percent of the energy in one level is transferred to the next level. The other 90 percent of the energy is used for normal life functions such as movement, growth, and reproduction, and to keep the body temperature constant in warm-blooded animals. The energy that stays in the organism is eventually released by decomposers after the organism dies. Eventually, this energy provides the energy needed for the growth of new organisms.

Figure 7.4 This energy pyramid demonstrates that it takes many organisms at lower levels to support fewer and fewer organisms at higher levels. The mouse that eats wheat gets only 10 percent of the energy the wheat captured from the Sun. The owl, the secondary consumer, gets only 10 percent of the energy the mouse got from the wheat.

Key Terms

chloroplasts
chlorophyll
photosynthesis
cellular respiration

Check Your Understanding

1. How do animals capture energy from the Sun?
2. What is cellular respiration? Why is it important for humans?
3. As you go higher on the pyramid of energy, is there more or less energy available? Why is this so?

7.2 The Need for Energy

The snow boarders in the photograph use several kinds of energy as they go down the hill. One major use of energy in our bodies is in maintaining **homeostasis**. No matter what is happening outside the body, homeostasis maintains the body's internal environment. Despite the chill of a cold Canadian winter, for example, human bodies try to hold a steady temperature of approximately 37°C.

Your body has many adaptations that allow it to stay at this temperature, regardless of the temperature of the surrounding environment.

- When your body is too warm, for example, blood vessels are brought close to the skin's surface. Heat can then transfer out quickly.
- Perspiration is another way the body cools. Moisture is moved outside the skin. Evaporation of this moisture cools the body.
- Shivering helps to warm your body. Vibrating muscles burn more energy. The chemical potential energy in glucose is converted to the kinetic energy used in shivering, and thermal energy is released.
- To protect itself from getting too cold, the human body can direct less blood to limbs and more to the core. To do this, blood vessels near the surface constrict. Less blood can get close to the outside, so less thermal energy is lost to the environment.

The rate at which a human body uses energy is referred to as **metabolism**. Someone with high metabolism has high energy needs. These people may be able to eat a lot but still not gain weight. That is because the energy from their food is either used by the body during digestion, activity, and excretion (getting rid of wastes), or is released, usually as thermal energy. People with low metabolism have lower energy needs.



Figure 7.5 Think of the types of energy you use when snowboarding. Muscular energy keeps you balanced on the snowboard. Energy used by your brain allows you to think quickly to avoid falls and other accidents.



The human body gives off about 100 calories of heat per hour. This is a little more than the heat given off by a 100 watt electric light bulb in one hour.

DidYouKnow?

A fever that raises body temperature too high (for example, over 42°C) can cause organ damage. It is particularly important to monitor and treat fever in infants and small children.

INVESTIGATION 7-B

Daily Temperature Changes



Hour	Temperature	Hour	Temperature
1	36.6	13	37.2
2	36.4	14	37.4
3	36.3	15	37.0
4	36.0	16	36.8
5	35.7	17	37.8
6	35.7	18	37.8
7	36.0	19	37.4
8	36.6	20	37.0
9	37.0	21	37.0
10	36.6	22	36.9
11	36.6	23	36.8
12	36.8	24	36.8

Think About It

Your temperature changes constantly throughout each day. Normal body temperature for a healthy individual is between 36.2°C and 37.2°C . But the body temperature of a healthy person can fluctuate as much as one or two degrees. When a person is sleeping, their body temperature is usually coolest between 4:00 A.M. and 5:00 A.M. The body is warmest during hard physical exercise.

What to Do

1 Use the information in the table to draw a line graph.

- The rectangle for the graph should be 26 units long and 25 units high.
- Place hours on the horizontal axis (x -axis).
- Make the vertical axis (y -axis) from 35.5°C to 38.0°C . Allow space for every tenth degree.

Analyze

1. At which hour or hours does the lowest temperature occur? What is this person probably doing at this time?
2. Examine your graph. At which point do you think this student is participating in a rigorous sport? Explain your answer.
3. (a) Assume the lowest temperatures begin at 4:00 A.M. Develop a time line below the horizontal axis (x -axis) that shows at which time of day each temperature was taken.
(b) Is there a relationship between normal eating times and temperature? What do you think could account for these changes?

Key Terms

homeostasis
metabolism

Check Your Understanding

1. How do homeostasis and metabolism work together to keep an organism alive?
2. List and explain four methods that an organism such as a cat uses to stay warm on cooler days.
3. Many exercise and weight-loss programs encourage speeding up an individual's metabolism. Explain how a person could lose weight by increasing their exercise routine.

7.3 Sources of Energy

Food is the fuel that keeps humans warm, active, and alive. The main source of energy for the human body is a simple **carbohydrate** called **glucose**. Through cellular respiration, glucose is broken down to provide energy that powers the body's metabolism. If more glucose is taken in than is immediately needed, the extra is converted to **glycogen**, a starch-like substance. Glycogen is stored in the liver and muscles. When we need energy, glycogen is converted back into glucose.

Health experts recommend that we take in 55 to 65 percent of our daily calories in carbohydrates.

People who go on special diets tend to alter the amount of carbohydrate, fat, and protein they eat because these nutrients provide the energy that keeps our bodies going. **Carbohydrate is the most direct source of energy.**

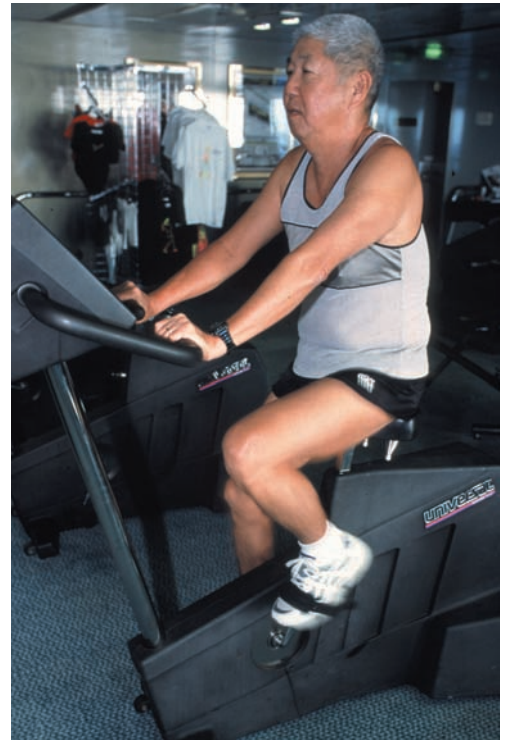


Figure 7.6 Regardless of your age or build, exercise will increase your metabolism. This increase can be measured by the amount of heat the body produces due to increased chemical reactions. Even after exercise, metabolism stays high for a short period.

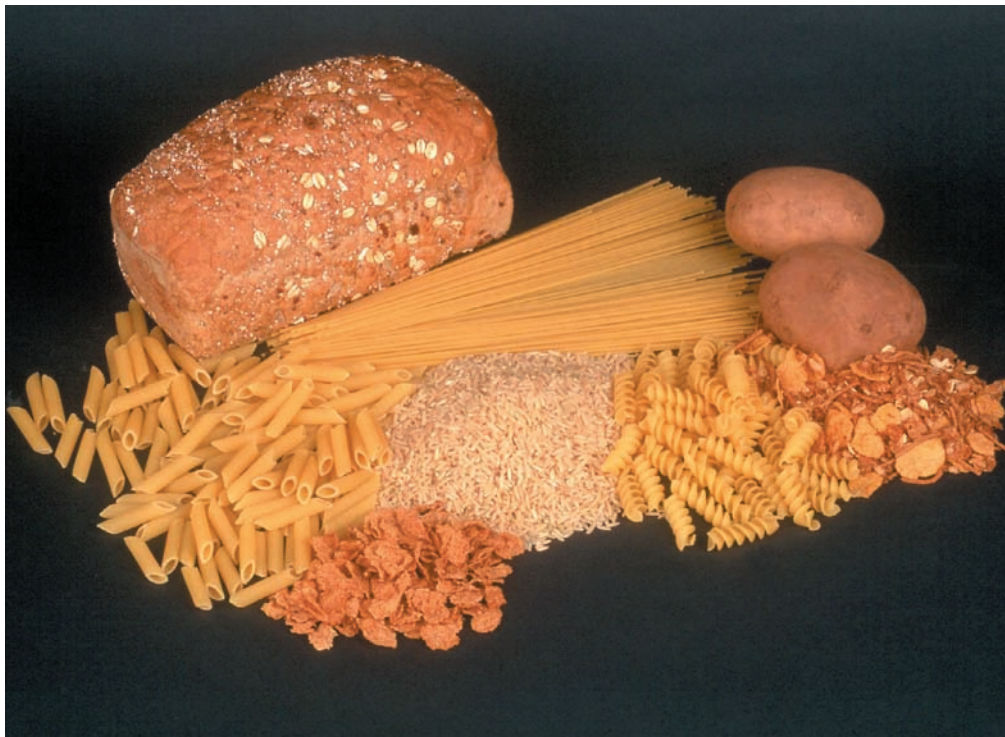


Figure 7.7 Pasta, bread, and potatoes are rich in carbohydrates. These carbohydrate-rich foods are your body's preferred source of energy. **When you need energy, your body uses carbohydrates first.**

DidYouKnow?

Want to increase your metabolism? Then build muscle. People with a lot of muscle have a higher metabolism because their body keeps all the muscle cells fuelled with energy.



Figure 7.8 Fat is often thought of in a negative way. This is unfair. Too much fat can be a problem, but bodies need fat for good health.

Fat: There When You Need It

We need body fat for insulation, and as a source of energy. Once carbohydrate stores of energy run low, the body turns to fat to power its activities. Kilogram for kilogram, **fat delivers almost twice as much energy as carbohydrates do.**

The **fats** we eat are classified in two groups: saturated and unsaturated. **Saturated fats** come from animal fats and from some oils. These types of fats are associated with an increased risk of heart disease. **Unsaturated fats** include most vegetable oils, such as Canola™ and olive. These fats are associated with reducing the risk of heart disease.

Dietitians recommend we have a fat intake of less than 30 percent of our daily calorie intake. They also recommend replacing saturated fats with unsaturated alternatives where possible.

READING check

Why is fat necessary?
What are the safest sources of fat?

Figure 7.9 These long-distance runners are training for a marathon, which is a 42 km run.



Protein: The Last Choice

One gram of **protein** contains the same energy as one gram of carbohydrate. The difference is that it is harder for the body to break it down to release the energy, so it takes longer. That is why the body will use protein as an energy source only when there is little carbohydrate or fat left. Protein provides about the same energy per kilogram as carbohydrates, and about half the energy provided by fat.

Although it is a poor source of energy, protein is important for other body functions. **Proteins help build and maintain body tissues. They are used for growth. That is why children, pregnant women, and women who are breast-feeding need larger amounts of protein.**

When proteins are used as an energy source, however, they are unable to perform their growth and repair functions effectively. So it is important to eat proper quantities of carbohydrate and fat.

10 percent of our daily diet should be proteins.



Figure 7.10 Body tissue and muscles are made of protein.

READING
check ✓

Why are proteins important in our diet?

Building Blocks

Proteins in our body are like letters of the alphabet. Letters are used to create words — they are the building blocks of our language. Proteins are the building blocks of our physical body.

Many people take proteins into their bodies by eating meat and other protein-rich foods. Most meat contains protein, which is necessary for growth and cell repair. **Vegetarians do not eat meat. They must be careful that the fruits and vegetables they do eat provide them with essential protein material.**

Some individuals choose extreme diets that can lead to health problems. They may severely limit food intake because of an eating disorder or to keep their weight down. This is more common in careers such as modelling, or in sports such as figure skating or gymnastics. The effects on the health and development of these people can be similar to those experienced by young people living in countries where food is very scarce.



Figure 7.11 This child does not get enough of any of the three essential nutrients. If she continues to be malnourished for much longer, she will be affected for life. Good nutrition later will not be able to repair the damage done during her development.

READING
check ✓

Why do children need more protein than adults?

7.4 Human Energy Needs Change



Figure 7.12 The developing fetus is dependent on the mother. At this point in life, the fetus gets enough energy if the mother eats well. Pregnant women should check with a dietitian to make sure they are eating the proper amount and variety of foods.

Our need for energy changes as we go through the many stages of life, from infancy through puberty to old age.

In the picture of the developing fetus, there is a long tube called the umbilical cord. This is the lifeline for the developing child. Nutrients from the mother pass through this tube to the developing fetus.

After birth, babies are still dependent on their mothers, or another caregiver. The best nutrition for babies is breast milk. Babies can thrive on an artificial milk formula that simulates the nutrients in breast milk. The milk formula contains the necessary nutrients and minerals that give babies the energy they need to grow.

As people age, their need for energy — called their **metabolic needs** — changes. Every growth spurt and every development has its own demand for energy. For example, young people require large amounts of energy during puberty, when they increase in height, increase in mass, and mature sexually and physically. Breast-feeding mothers have high metabolism because they are providing nutrients both for their own bodies and that of the child or children they are breast-feeding.

READING Check

Explain why teenagers need more energy than adults.

DidYouKnow?

Babies who are breast-fed receive antibodies from their mother. These antibodies help to strengthen the child's immune system. Studies have shown that breast-fed babies are less likely to become sick.

Career CONNECT

Fitness instructors coach people in aerobics, weight training, and flexibility. Some also provide counselling about proper nutrition and weight control.

There is a growing interest in health and fitness in Canada. If you are physically fit, have a good sense of rhythm, speak well, and are interested in helping others live and eat well, this might be a good career for you.

An instructor usually needs training and certification from a fitness association. Each province and territory has its own set of requirements. To learn about the requirements where you want to work, call the National Fitness Leadership Advisory Council or talk to employers such as the YMCA and YWCA. Many facilities offer in-house training and certification.



Figure 7.13 As well as helping clients through exercise routines, fitness instructors give advice on proper diet and the best ways to control weight.

What Affects Metabolism?

Age —Developing fetuses and young children have much faster metabolisms than adults.

Time of day — Metabolism is higher in the morning. It comes to a peak around 10:00 A.M., and slows throughout the day.

Exercise — Activity raises metabolism. The best way to lose weight is to maintain the same food intake but increase the amount of exercise.

Level of fitness — Athletes and others who are physically fit have a high metabolism because they use up large amounts of energy getting and keeping in shape.

One way to avoid gaining weight when metabolism drops is to keep active. Take long walks, swim, and cycle. Another way is to reduce food intake.

When people cut back on their food volume, they need to reconsider their dietary habits. For example, high amounts of fatty foods may increase the chance of heart disease; lack of calcium can cause brittle bones.



Figure 7.14 Thermal energy is released as the body converts chemical potential energy to the kinetic energy of motion during exercise. As a result, active people have a faster metabolism and a higher need for energy. Which of the people in this photo likely has the highest metabolism?

READING check

List the factors that affect metabolism. Explain what effect each factor has.

Check Your Understanding

1. In your own words, define “metabolism.”
2. Describe some of the differences in nutrient and energy requirements between a baby and a senior citizen.
3. List three occupations in which you would expect workers to have high metabolisms.
4. For each of the following situations, predict whether the person’s metabolic rate will go up, down, or stay the same.
 - (a) a construction worker resigns and becomes a bus driver.
 - (b) a mother stops breast-feeding
 - (c) a high-school gymnast begins training for the provincial finals

Key Terms

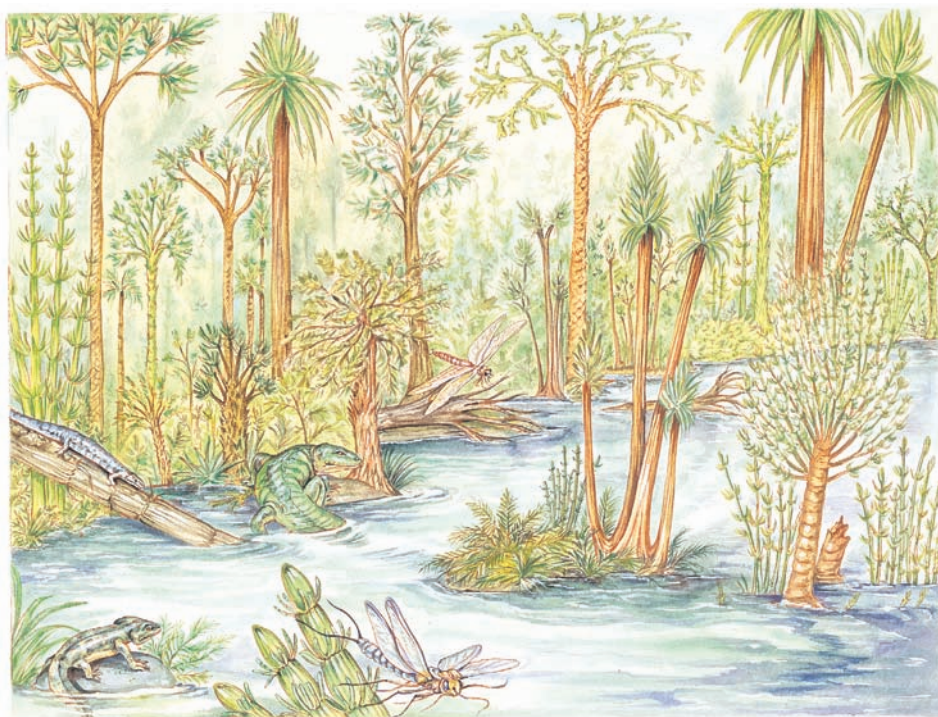
metabolic needs

8

Fossil Fuels

Getting Ready...

- List the fossil fuels that you know of.
- What do you already know about how fossil fuels were formed?
- What are the risks of burning fossil fuels?
- How is the fossil fuel industry important to Alberta?



Imagine what Alberta must have been like millions of years ago. Temperatures were much warmer. Lush green vegetation covered much of the land. The rest was covered by oceans.

Abundant life filled the ocean and forests. When organisms died, their remains accumulated in stagnant swamps. Over centuries, seas rose or land sank, and the swamps were covered.

Solar energy captured by those ancient plants and animals has been transformed and is still here for us to use. How did these incredible transformations take place?

Today our society relies heavily on the remains of ancient forests and long-dead marine life. From their remains, we produce gasoline and diesel to fuel motor vehicles. We transform other energy stored as coal and natural gas into electric energy.

In this chapter you will learn about the formation of fossil fuels. You will examine their extraction and refining, and you will investigate their burning. Finally, you will analyze the impact that burning fossil fuels has on the environment.



What You Will Learn

In this chapter you will learn:

- when fossil fuels were formed
- how fossil fuels are extracted and refined
- how combustion of fossil fuels and cellular respiration are related
- about the risks and benefits associated with the use of fossil fuels

Why It Is Important

- Energy from fossil fuels powers vehicles for businesses and individuals. For most of us, this fuel heats and lights our homes and cooks our food. By learning about the positive and negative aspects of fossil fuels, you will be able to make better judgements about how you use them.

Skills You Will Use

In this chapter you will:

- model what happens to oil, water, and gas under the Earth's surface
- model the distillation of crude oil
- investigate the combustion of fossil fuels
- analyze the impact of fossil fuel technologies on the environment

Starting Point



Oil and Water

Not long ago, you could not buy bottled drinking water in Canada. How much does it cost today? How much is oil worth? Which of these liquids is the most important to *your* life?

What to Do



1. Use newspapers, advertisements, a web search, or other sources of information to determine the following:
 - (a) What is the price, in your area, for one litre of motor oil?
 - (b) What is the price, in your area, for one litre of bottled drinking water?
 - (c) At our current rate of use, how long do scientists believe our global oil reserves will last? Are we using more or less oil than 20 years ago?
 - (d) Is Canada in danger of facing water supply shortages? Should we be building pipelines to transport water to the United States? Should we ship it to other countries? Why or why not?

8.1 What Are Fossil Fuels?



Figure 8.1 This fish lived millions of years ago. You can still see its shape in this rock.

When you hear the word “fossil,” you may think of the skeleton of a fish embedded in rock, such as the one shown in Figure 8.1. Or you may think of shell-like impressions in rock. With this knowledge you are well on your way to understanding **fossil fuels**.

Coal, crude oil, and natural gas are commonly used fossil fuels. Like the fossils you can see in rock, they were once living organisms. Instead of being converted into fossils, the organic material from these plants and animals piled up on the bottom of a swamp, lake, or ocean. It was then covered with many layers of sediment.

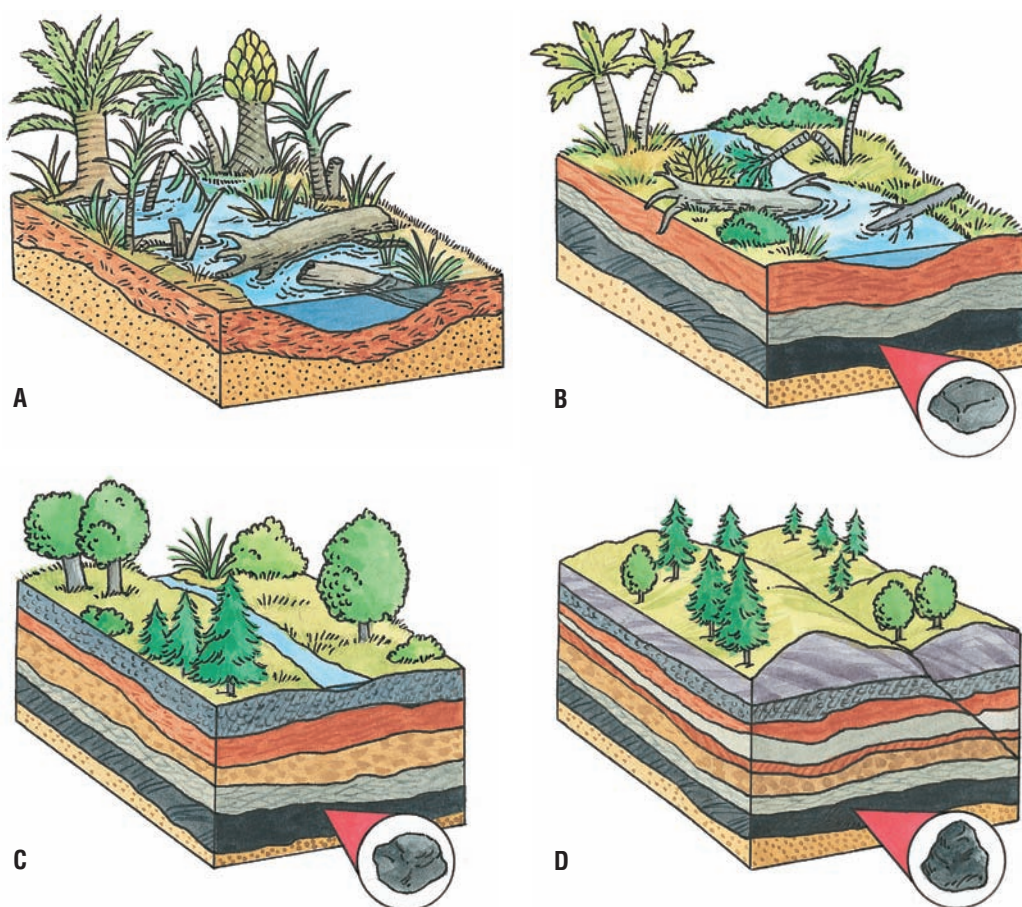


Figure 8.2 Fossil fuels take a long time to develop. These visuals show how different types of coal developed. Crude oil was formed in a similar way.

- A** Coal was formed from the remains of ancient swamp vegetation that lived about 300 million years ago.
- B** The plants died and fell into the swamp. Their organic material was covered by layers of sediment. There was no oxygen in these layers, so the plant material could not decompose. Hydrogen and carbon from the organic material were locked together underground. Eventually, they formed a solid layer of **lignite**, which is a soft coal.
- C** More layers of sediment piled on top of the lignite and compressed it even further. Because the layers were so deep and there was so much pressure, temperatures increased. The heat and pressure compressed the lignite into a harder coal, called **bituminous coal**.
- D** In some areas, layers of bituminous coal were compressed and heated even further. This produced an even harder coal called **anthracite**.

The layers of sediment were often several kilometres thick. Heat and pressure changed the decaying organic matter into **coal**, oil, and natural gas. Because they formed from plants and animals that lived long ago, we call these three materials fossil fuels.

The Use of Coal, Oil, and Natural Gas

Coal has been used as a fuel for at least 3000 years. It is abundant in most parts of the world, but was used only a little until the invention of the steam engine. For more than 100 years, steam engines were used to power factories and move people and goods.

Coal was and still is popular because it is cheap, easy to mine, and easy to transport. That is why it provided almost all of the thermal energy for the steam engines that made our country grow — and why it is still popular today for some industrial uses.

Oil and gas have not been used for as long a period as coal. Ancient eternal fires burning in the Iraq area were reported by travellers 1900 years ago. These were probably the result of **natural gas** escaping from cracks in the ground and being ignited by lightning.

Natural gas is difficult and dangerous to control and transport. That is why its potential energy has only been used in recent years.

Crude oil, also called **petroleum**, is found as a liquid. Crude oil is extracted by drilling a hole into Earth's crust. In some cases, the oil is under pressure and will come out of the hole on its own. In other locations, it has to be pumped to the surface.

DidYouKnow?

There are three types of coal found in Alberta.

Lignite is a soft brown or black coal found in southeastern Alberta.

Sub-bituminous coal is softer than **bituminous** and contains more moisture. Alberta is the only province in Canada where **sub-bituminous** coal is mined. It is used to generate most of the province's electricity.

Bituminous coal is also found in many areas of the province.



Figure 8.3 Tiny diamonds fixed in the tip of drills cut holes through rock to oil and gas reservoirs.

Key Terms

fossil fuels
coal
natural gas
crude oil
petroleum

Check Your Understanding

1. List three fossil fuels.
2. Sketch and label four steps in the formation of coal.

8.2 Extracting Coal and Oil

Since the late 1800s, the coal industry has been an important part of Alberta's history.

READING check

List two ways Alberta coal is used.

- For more than a century, coal was the main energy source for the Canadian Pacific Railway and most of Alberta's industries.
- Most of Alberta's electric energy is produced in coal-burning thermal plants.
- Most of the iron and steel used in Alberta's industries has been produced in other parts of the world. Many of these other areas use the potential energy stored in coal from Alberta to produce iron and steel.

Mining Coal

How is coal mined for all these purposes? The coal formed during the process shown in Figure 8.2 developed into flat layers. Depending on the area, these layers occur as thick beds or long seams. Over many millions of years, these beds and seams of rock were folded and deformed. In order to mine coal, miners look for a seam or bed of coal.

Deposits close to the surface of Earth are often quarried in much the same way as sand and gravel are quarried. Coal quarries are called open-pit mines.

Deposits that are deep below Earth's surface are dug using networks of underground rooms and tunnels. Machinery cuts into the coal, and conveyor belts lift it to the surface.



Figure 8.4 Coal discoveries in western Canada aided and encouraged settlement by Europeans.

READING check

Explain the difference between an underground and an open-pit mine.



Figure 8.5 An open-pit mine devastates the landscape. Trees and soil are removed to retrieve coal. Environmentally conscious coal companies now reclaim the land by covering the area with topsoil. They grade the area to form a contoured landscape, then seed the area with native plants such as grasses and trees. Notice where reclamation has been started along the edge of this open-pit mine in Germany.

Oil

A hundred years ago crude oil was used to make kerosene, which is a fuel for lamps and stoves. Today crude oil is the world's most important source of energy.

In Alberta, the drilling for crude oil dates back to 1914. In that year, a miner named W.S. Heron noticed oil seeping from the ground on a farm near **Turner Valley**. He purchased the land and mining rights, and started drilling.

READING
check ✓

Why is natural gas found on top of water and oil?

This discovery of oil near Turner Valley, and later at Leduc, Woodbend, and Redwater, has made Alberta into a major oil-producing area. Today, the province exports energy to the rest of Canada and the United States.

One problem with mining crude oil is knowing where to drill. Look at Figure 8.7 to see what a difference picking the right place to drill can make!

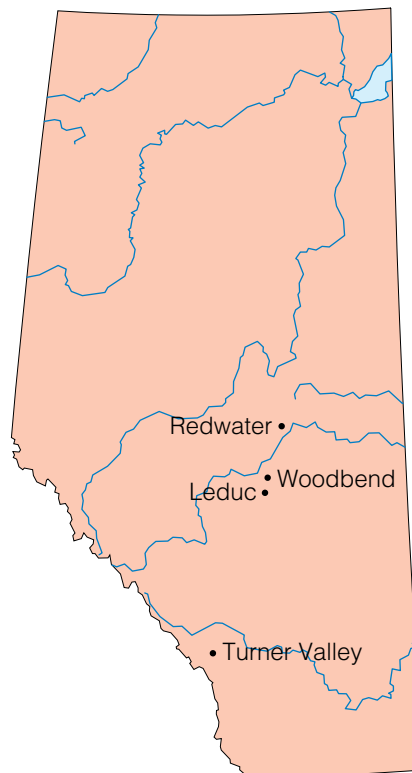


Figure 8.6 Alberta's Early Oil Deposits Research to find the areas of Alberta where oil is produced today. How would the loss of this industry affect Alberta's economy?

Searching for Oil

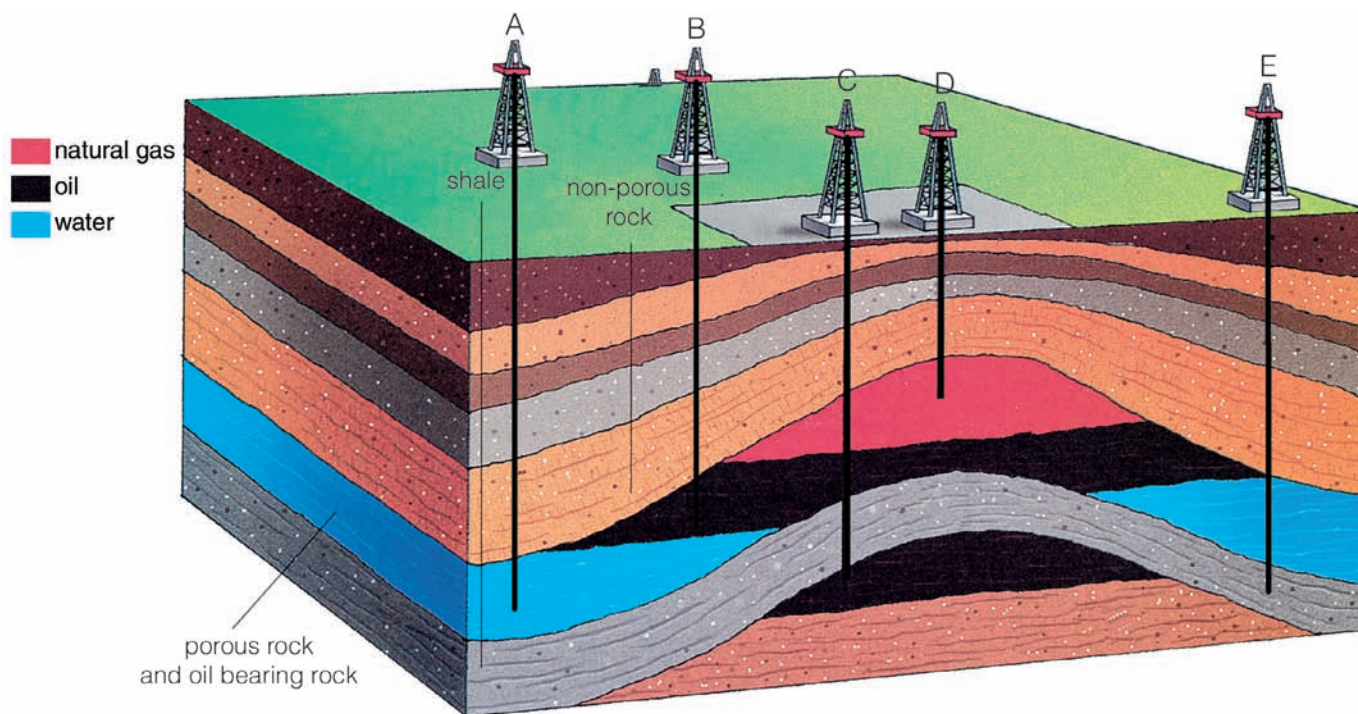


Figure 8.7 As oil developed, it mixed with water and sand. This mixture, along with bubbles of natural gas, seeped through porous layers of rock. Eventually, the mixture reached a layer of rock it could not pass through and was trapped.

The water, oil, and natural gas gathered under the non-porous layer of rock. Because natural gas is less dense than oil, it rose to the top.

The well at D pumps natural gas. What do the wells at A, B, C, and E get?

Diagrams such as Figure 8.7 may give the impression that oil forms in large, underground lakes. This is not correct. Oil is found in microscopic pores between the particles that make up rocks.

Over time, changes in the Earth's crust caused great amounts of pressure that forced the tiny oil droplets into porous rock called **reservoir rock**. Further movements of the Earth's crust helped to trap oil and natural gas in sections of reservoir rock that are located between layers of impermeable rock. Figure 8.7 shows an oil-bearing structure that was formed as a result of these processes.

READING check

Explain how geologists use a seismic survey.



Figure 8.8 In this print out from a seismic survey, the different layers of rock are clearly visible.

Geologists use a number of technologies to determine the location of potential oil and natural gas reservoirs. One of the more common methods is the use of a **seismic survey**.

During a seismic survey, shock waves from compressed air guns, thumper trucks, or explosives are sent through the layers of the Earth's crust. The returning sound waves reflect off rock. They are detected by sensitive equipment and used to create images of the underground rock layers.

Geologists interpret these images and other data looking for structures that may contain oil and natural gas.



Drilling for Oil

Once seismic surveys identify likely places to find oil and gas, oil companies start to drill. When they hit reservoir rock with oil or gas, the fuel may come to the surface on its own.

As oil is removed from the reservoir, however, the pressure of the oil drops. In these cases, a pump is installed to bring the oil to the surface.

When is a pump used to bring oil to the surface?

DidYouKnow?

A garden hose with a hand-controlled nozzle can be compared to an oil well. The water in the hose is like the reservoir rock. Before the rock has been drilled into, the oil is under a lot of pressure.

Opening the nozzle is like drilling into the reservoir rock. In the case of the hose, water shoots out the nozzle. In the case of reservoir rock, oil may shoot into the air if it is under a lot of pressure. Otherwise, oil will be pumped to the surface.



Pump Jack

Refining Oil

Once crude oil is extracted from the ground, it is refined into many different products. This is done by a process called **distillation**. Distilling allows people to separate materials with a low boiling point from those with a higher boiling point. When oil is refined, it goes into a distillation system. Here, heavy lubricants such as motor oil and kerosene are separated from lighter liquids such as gasoline and naphtha.

Natural Gas

In Figure 8.7 you learned that natural gas is found underground in rock formations. You can see in the figure why it is often discovered in the same location as crude oil.

Until a few decades ago, when natural gas was found while drilling for oil it was simply burned off and wasted. Back then, natural gas was difficult to control. Oil was easier to control and transport.

Today there are pipelines across Canada to transport natural gas. In addition, trucks and ships have been developed that can transport it safely. As a result natural gas is used to heat houses and run industries.

Natural-gas pipelines run to towns and cities thousands of miles from the wells. These deliver natural gas directly to consumers, including homes, schools, and factories. Natural gas is a popular fuel because it causes less pollution than either coal or gasoline. In addition, it provides an immediate heat.

Like other fossil fuels, when natural gas is burned the stored energy is transformed to thermal energy for heating and cooking. The energy in natural gas can also be used to generate electric energy and provide thermal energy needed in some factories.



Figure 8.9 Old Glory was the name of the discovery well drilled by Eugene Coste. This well was part of the **Bow Island gas field**, which was the first major discovery of gas in Alberta. The development of this field led to the first natural-gas pipelines to deliver gas to Alberta communities.

Check Your Understanding

1. Describe two ways in which coal is mined.
2. List three uses for each of the three fossil fuels.
3. Explain how the various components of crude oil can be separated.
4. Use a graphic organizer to show how oil, water, and natural gas arrange themselves under Earth's surface.
5. Explain how shock waves are used to find oil.



How is natural gas used?

Key Terms

reservoir rock
seismic survey
pump jack
distillation

Combustion



Figure 8.10 A blue flame is a sign of complete combustion. Smoky yellow flames indicate incomplete combustion. They are a sign that the flame is not getting enough oxygen.

READING check

What two products are formed when a fossil fuel completely combusts?

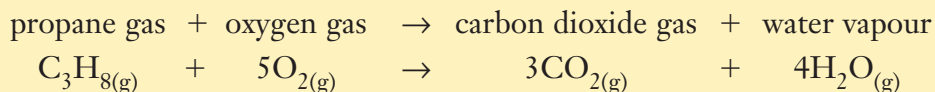
Energy stored in fossil fuels is converted to thermal energy by **combustion**. Combustion is a chemical reaction that takes place when a substance combines with oxygen.

In the case of fossil fuels, combustion happens quickly. During the reaction, thermal energy is released.

Many people use propane when they want immediate heat. Propane is one of several gases that can be separated from natural gas. Combustion of this gas provides thermal energy to heat houses, as well as cook foods in restaurants and on barbecues. It is also used to run some motor vehicles.

The chemical symbol for propane is $C_3H_{8(g)}$. When propane burns completely, the equation for the reaction looks like this:

Combustion of Propane



Like propane, all fossil fuels are made up of carbon and hydrogen. For this reason they are often known collectively as **hydrocarbons**. When fossil fuels burn completely in the presence of oxygen, they produce carbon dioxide gas and water vapour. Complete combustion ensures the maximum amount of thermal energy. It also ensures that carbon dioxide is produced.

If combustion is not complete, carbon monoxide will also be produced. Unlike carbon dioxide, this is a poisonous gas. Study Figure 8.8 to find out how you can tell whether a gas flame is producing carbon dioxide ($CO_{2(g)}$) or carbon monoxide ($CO_{(g)}$).

DidYouKnow?

It is dangerous to use a gas barbecue or an unvented propane heater indoors. There is less oxygen available indoors than there is outside. When there is little oxygen, combustion is not complete and deadly carbon monoxide gas can be produced.

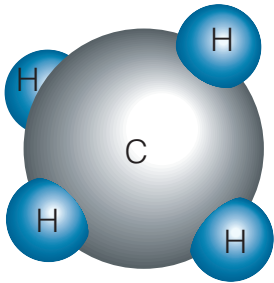
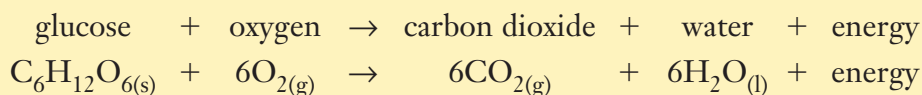


Figure 8.11 This artist's idea of a methane molecule shows four hydrogen atoms joined to a carbon atom. At the beginning of this section you saw the chemical formula for propane. Use this model to help you sketch a propane molecule.

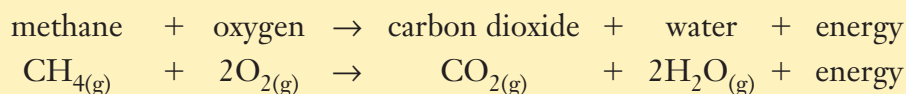
Combustion and Cellular Respiration

Cellular respiration is a process similar to combustion. **Study the two equations below to see how these two processes are similar.**

Cellular Respiration



Combustion of Methane



1. What “fuel” is burned in each case?
2. What gas does each source of energy react with?
3. What are the products of these reactions?

Your answer to questions 1 to 3 should show you that cellular respiration and the combustion of methane are similar chemical reactions. **Both reactions produce a source of energy and use the gas oxygen. Carbon dioxide and water or water vapour are products of both.**

A difference between these reactions is what happens to the energy. In combustion reactions, thermal energy is released. In cellular respiration, much of the energy is used to produce chemical energy for the body. Only a small amount is transformed into thermal energy. Another major difference is the rate at which the two reactions occur. Combustion happens rapidly, while respiration is slow and controlled.

Key Terms

combustion
hydrocarbons

Check Your Understanding

1. List three technologies that use combustion to convert the energy stored in fossil fuels into other forms of energy.
2. Copy the word and chemical equations for the combustion of propane ($\text{C}_3\text{H}_{8(g)}$). Label the input energy, converter, and output energy.

8.4 The Other Side of the Coin

We need fossil fuels to keep us warm, to keep our homes and streets lighted, and to run our industries. Alberta's good supply of fossil fuels has a direct impact on the economy of the province.

READING
check ✓

Why are fossil fuels said to be non-renewable?

Unfortunately, there is another side to this coin. **Fossil fuels are non-renewable.** Once used, they are gone. They cannot be re-used and will not form again within the foreseeable future. Just like fossil fuels from the past, plants and animals that die today will take hundreds of millions of years to become fossil fuels.

At the rate we are using these fuels today, we will run low in some fuels. As global population and demand increases, more and more people compete for less and less oil, gas, and coal.

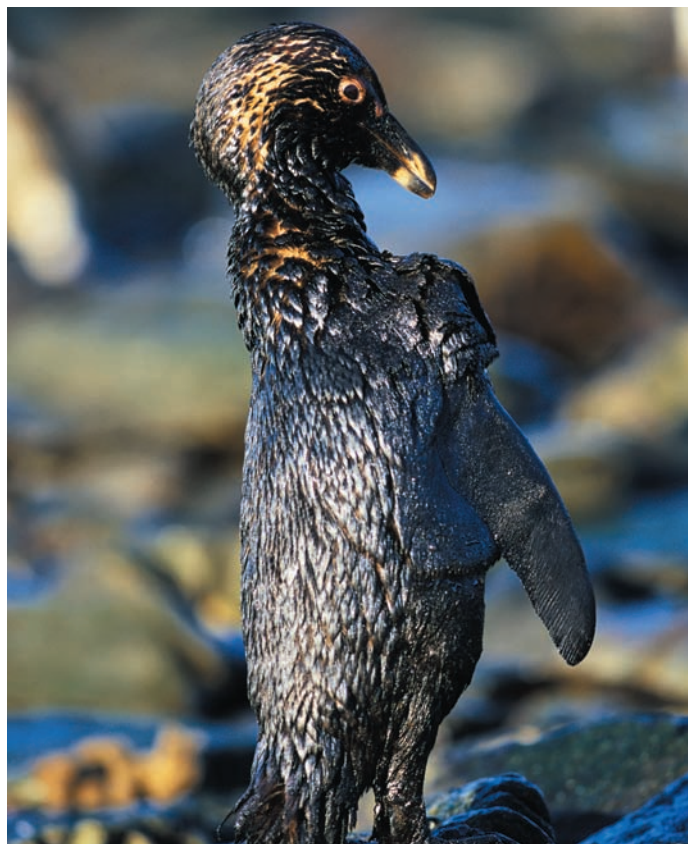


Figure 8.12 Fossil fuels take millions of years to accumulate. They are burned in minutes by internal combustion engines. As they combust, they create other problems. Which problem is evident in this photo of Vancouver?

READING
check ✓

Name two problems we have when oil tankers run aground or leak oil.

Figure 8.13 An oil spill puts marine life — fish, mammals, and birds — in danger. If they swallow the oil, it poisons them. If the oil covers their feathers or fur, it destroys natural insulation and waterproofing. The animals drown or freeze. This African penguin is covered with oil from a spill off the coast of South Africa. Some species of penguin are approaching extinction from this cause alone.



Acid deposition raises natural acidity in lakes and streams. When the acid levels get too high, fish cannot reproduce. Some lakes in northern Canada were once considered dead because acid deposition had killed the plants, algae, and fish that lived in them. Since the emissions from nearby industries have been cleaned with scrubbers, the lakes have begun to recover.

READING
check ✓

How does burning fossil fuels harm the environment?

Greenhouse Gas



Figure 8.14 The combustion of gasoline and diesel fuel releases unburned fuel, sulfur dioxide, nitrogen oxides, soot, and carbon dioxide into the atmosphere. This can result in smog, which is a combination of air pollution (smoke) and fog. Breathing air like this is bad for your health.

Unit A discusses the greenhouse effect. The natural greenhouse effect is good because it traps thermal energy and makes Earth warm enough for living things. But many scientists think the burning of fossil fuels has resulted in an increase in greenhouse gases. This higher level of greenhouse gases is trapping more heat inside Earth's atmosphere. The majority of scientists believe this will cause changes to our climate.

They predict that some places will get much warmer, while others will get cooler. Scientists call this phenomenon **global climate change**.

Aboriginal groups in northern Canada also say that the climate is changing. They report that winters are getting warmer and that the permafrost is melting. Scientists say that increasing temperatures and decreasing precipitation may lead to new deserts. In addition, they have found that Earth's ice caps and glaciers are melting faster today than they were a few decades ago. Melted glaciers will raise ocean levels and flood low-lying areas.

DidYouKnow?

Particles from fuel emissions, as well as nuclear wastes and PCBs, are being found in glaciers around the world, including those in the Rocky Mountains of Alberta.

Check Your Understanding

1. List three risks associated with our use of fossil fuels.

Key Terms

non-renewable
acid deposition
global climate change